



THE UNIVERSITY
OF QUEENSLAND
AUSTRALIA

CREATE CHANGE

Introduction to the course

DENT1020 Dental Science I

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Acknowledgement of Country

- The University of Queensland (UQ) acknowledges the Traditional Owners and their custodianship of the lands on which we meet.
- We pay our respects to their Ancestors and their descendants, who continue cultural and spiritual connections to Country.
- We recognise their valuable contributions to Australian and global society.

*The Brisbane River pattern from A Guidance Through Time
by Casey Coolwell and Kyra Mancktelow.*



Starting dentistry at UQ

How do you feel?



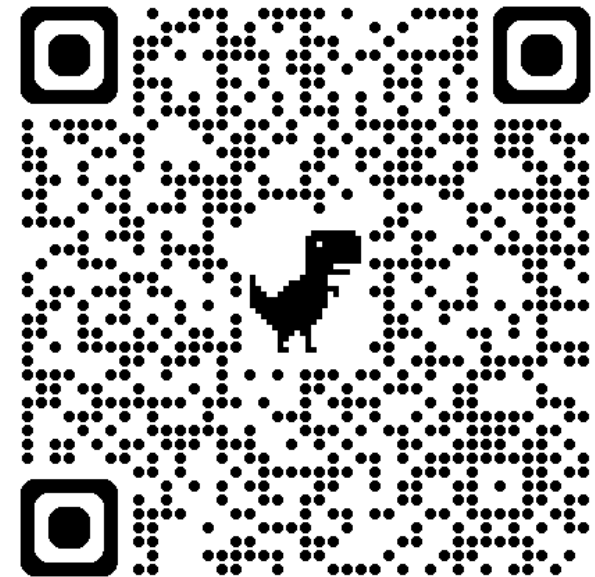


What does oral health mean to you?

Using 1-3 words, describe what oral health means to you

Class padlet wall:

<https://padletuq.padlet.org/sshetty7/class-padlet-wall>



For instructions on how to connect your wireless device to the UQ WiFi network go to: <https://its.uq.edu.au/services/connect-to-uq-wifi>

FDI's Definition of Oral Health



<https://www.youtube.com/watch?v=nzzzdy-k7KA&app=desktop>

Oral Health Definition: FDI

“Oral health is multi-faceted and includes the ability to speak, smile, smell, taste, touch, chew, swallow and convey a range of emotions through facial expressions with confidence and without pain, discomfort and disease of the craniofacial complex (head, face, and oral cavity).

Oral health means the health of the mouth. No matter what your age, oral health is vital to general health and well-being.”

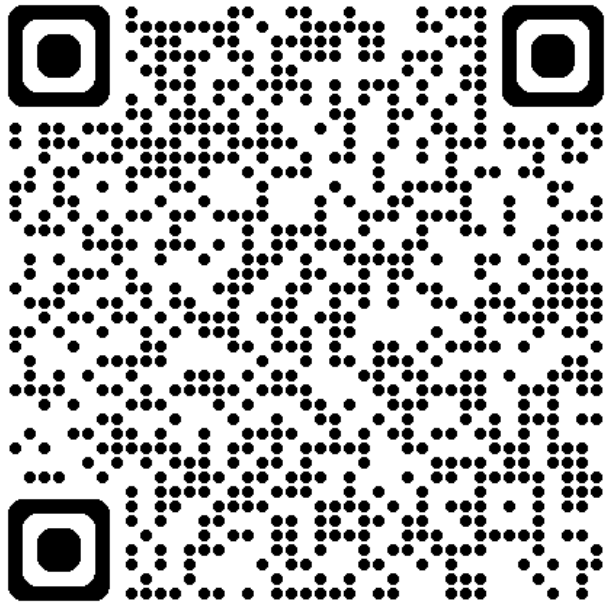


Oral health...

- ✓ **Fundamental component** of health and physical and mental well-being.
- ✓ Exists along a **continuum** influenced by the values and attitudes of individuals and communities
- ✓ Reflects the physiological, social and psychological attitudes that are **essential to the quality of life**;
- ✓ Influenced by the individual's **changing experiences, perceptions, expectations** and ability to adapt to circumstances.



Why did you choose to study dentistry?



<https://padletuq.padlet.org/sshetty7/2026>



This course introduces you to the underpinning science of dentistry and provides a foundation for understanding the normal structure and function of the oral cavity, the development and eruption of the dentition and maintaining a healthy oral cavity.

By working through a series of learning activities, students will develop a knowledge base that integrates a number of scientific disciplines, and an ability to apply this knowledge to the field of dentistry and evidence-based patient care.

Why?



Professional competencies of the newly qualified dental practitioner

May 2022 (effective 2023)



The structure of the statements

The range of competencies expected of the newly qualified practitioner has been clustered into the following six domains:

Competencies **for all divisions** of general registration:

1. Social responsibility and professionalism
2. Communication and leadership
3. Critical thinking

Competencies **specific to each division** of general registration:

4. Health promotion
5. Scientific and clinical knowledge
6. Person-centred care*

* has sub-domains of Clinical information gathering, Diagnosis and management planning, Clinical treatment and evaluation.

The domains represent the broad categories of professional activity and concerns that occur in the practice of dentistry.

As indicated above, there is a degree of artificiality in the classification, as effective professional performance requires the integration of multiple competencies.

Each domain contains descriptions of competencies. The descriptions are presented in one of two formats:

- Those descriptions for a **dental practitioner** are where the application of the knowledge and skills are the same for all divisions of registration.
- Those descriptions for **specific dental practitioners** that may be worded the same or in a similar manner, although the application of the knowledge and skills may vary between the different divisions of dental practitioner under the category of general registration.

Professional Competencies

Dentists

1. apply the social, cultural, biological, biomedical, physical and behavioural sciences in relation to oral health care provision and disease prevention
2. apply the theories and principles of population oral health
3. apply the scientific principles of infection prevention and control
4. understand the scientific basis, limitations, risks of dental materials and demonstrate their use
5. apply the principles of risk management and quality improvement
6. apply the principles of pharmacology, understanding the limitations and risks of using therapeutic agents, including polypharmacy and overuse, and the implication of the Prescribing Competencies Framework on dental practice
7. understand the scientific basis, risks, and demonstrate the safe use of ionising radiation

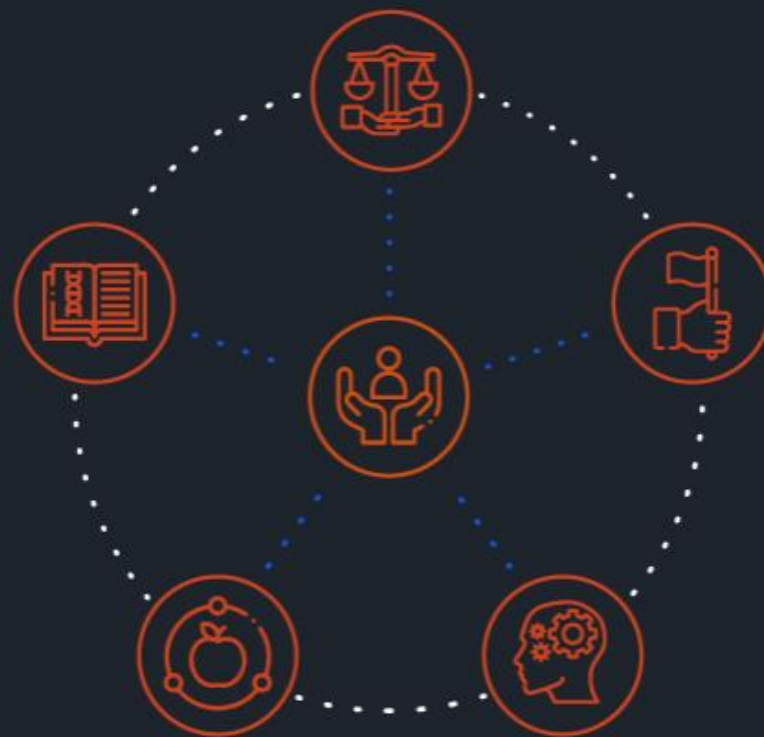


5. Scientific and clinical knowledge

Covers the application of the underlying knowledge required by dental practitioners

Professional competencies
of the newly qualified
dental practitioner

May 2022 (effective 2023)



Professional Competencies

Dentists

1. obtain and record a relevant history of the individual's medical, social, dietary, and oral health status
2. perform an examination for health, disease and abnormalities of the dentition, mouth and associated structures
3. select necessary clinical, pathology and other diagnostic procedures and interpret results
4. maintain accurate, consistent, legible and contemporaneous records of patient management and protect patient privacy
5. evaluate individual patient risk factors for oral disease
6. request and/or take radiographs relevant to dental practice

Dentists

1. recognise health as it relates to the individual, taking into consideration medical, social and cultural contexts
2. diagnose disease or abnormalities of the dentition, mouth and associated structures and identify conditions which require management
3. determine the impact of risk factors, systemic disease and medications on oral health and treatment planning
4. determine when and how to refer to the appropriate health and or care professional
5. obtain and record informed consent and financial consent for treatment
6. formulate and record a comprehensive person-centred, evidence-based oral health treatment plan



Professional competencies
of the newly qualified
dental practitioner

May 2022 (effective 2023)



6. Person-centred care

6.1 Clinical information gathering

Covers the collection and recording of information that is necessary and relevant



6. Person-centred care

6.2 Diagnosis and management planning

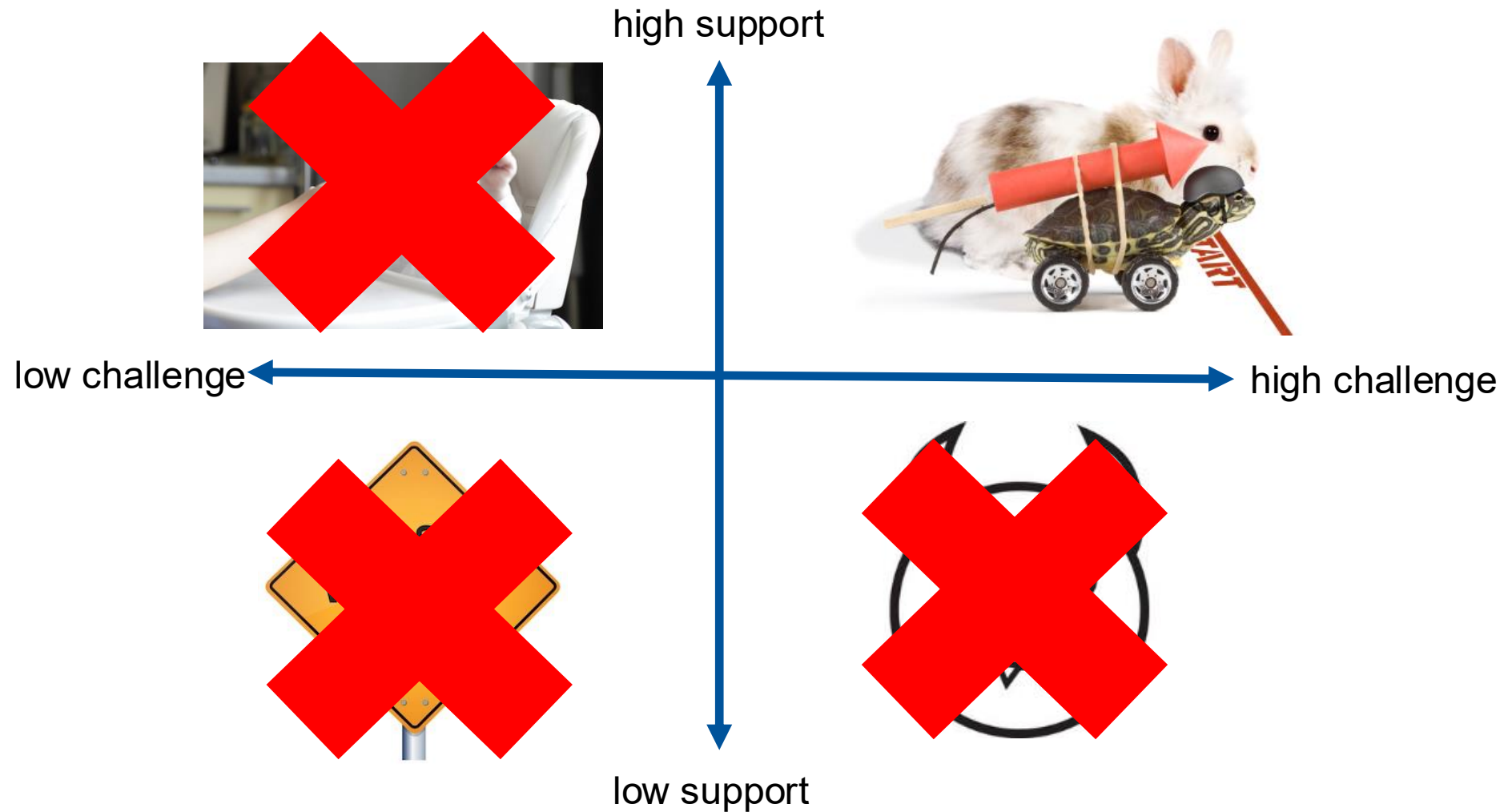
Covers the identification of disease or abnormalities that require treatment or investigation



The BDSc(Hons) program at a glance

Year 1	Semester 1	DENT1060 (2 units) Dental Practice IA	DENT1020 (4 units) Dental Science I	BIOM1050 (2 units) Introductory Cellular Physiology	
	Semester 2	DENT1061 (2 units) Dental Practice IB	CHEM1222 (2 units) Chemistry for Dentistry	BIOM1051 (2 units) Integrated Anatomy and Physiology	HEALTH1000 (2 Units) Profession, People and Healthcare
Year 2	Semester 1	DENT2044 (2 units) Immunology & Microbiology for Dentistry	DENT2060 (2 units) Dental Practice IIA	DENT2070 (2 units) Introduction to Dental Disciplines A	BIOM2052 (2 Units) Integrated Physiology & Pharmacology for Dentistry Dr Prasad Chunduri
	Semester 2	DENT2061 (2 units) Dental Practice IIB	DENT2071 (2 units) Intro to Dental Disciplines B	DENT2074 (2 units) Head & Neck Anatomy for Dentistry	DENT2222 (2 units) Applied Oral Biosciences
Year 3	Semester 1	DENT3000 (2 units) Population Oral Health & Professional Practice I	DENT3060 (2 units) Dental Practice III A	DENT3070 (4 units) Dental Disciplines B	
	Semester 2	DENT3021 (2 units) Research Informed Practice I	DENT3061 (2 units) Dental Practice III B	DENT3071 (4 units) Dental Disciplines B	
Year 4	Semester 1	DENT4021 (2 units) Research Informed Practice II	DENT4060 (2 units) Dental Practice IVA	DENT4070 (4 units) Advanced Dental Disciplines A	
	Semester 2	DENT4000 (2 units) Population Oral Health & Professional Practice II	DENT4061 (2 units) Dental Practice IVB	DENT4071 (4 units) Advanced Dental Disciplines B	
Year 5	Semester 1 <i>Course coordinator</i>	DENT5023 (2 units) Research Informed Practice III	DENT5100 (7 units) Dental Practice VA		
	Semester 2		DENT5200 (7 units) Dental Practice VB		

What can you expect from us?



What are our expectations of you?

Select a silent environment and turn off your voice during lectures

Silent mobile phones in class

Respect for me (no Facebook, no emails, no web browsing during class)

Respect for class mates (no judgement)

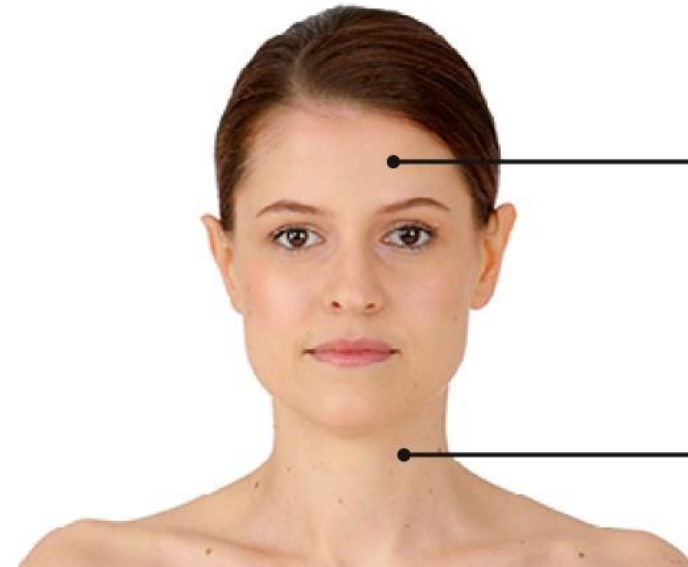
Active participation in learning activities

Communication policy

- All non-personal messages should be posted to the Blackboard discussion boards.
- I will check Blackboard regularly and will endeavour to answer any unanswered queries within 24-72 hours.
- Personal messages should be directed to the course coordinator with DENT1020 in addition to the subject in the subject line. Without this, the email may get lost or delayed.
- Check your UQ email daily. Communications sent to UQ email addresses will be considered 'received' within 24 hours of sending.

After successfully completing this course you should be able to:

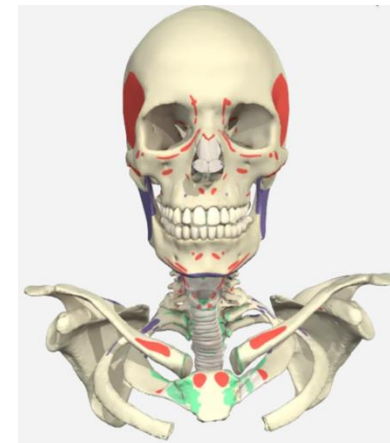
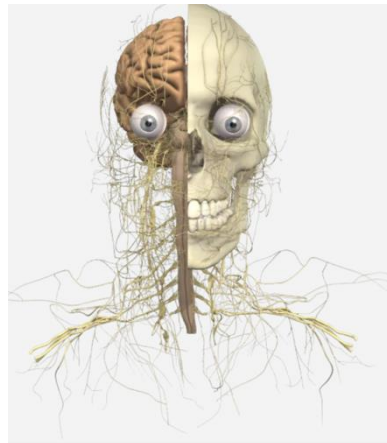
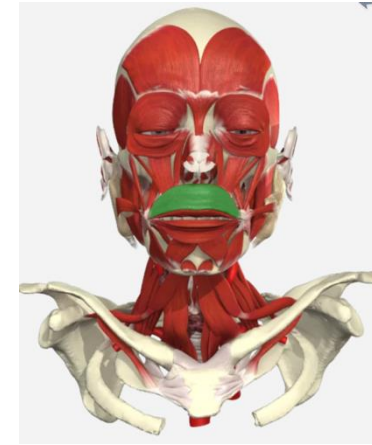
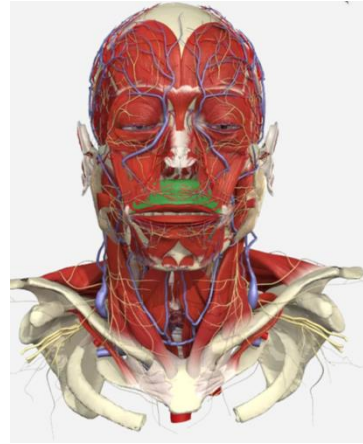
1. Describe in detail the anatomy and histology of structures within and around the oral cavity, including their clinical appearance using appropriate terms.
2. Demonstrate an understanding of the basic biologic sciences relevant to dentistry, including embryology at a fundamental level as well as normal tooth development and eruption.
3. Understand the role of both microbes and saliva in the oral environment, in both health and common dental diseases and conditions.
4. Demonstrate an understanding of the basic physical and chemical sciences relevant to dentistry, including physical forces and acid-base and solubility equilibria.
5. Use contemporary information technologies to critically identify and evaluate scientific evidence while contributing effectively to collaborative teamwork.



Head

Neck

Course Learning Objectives



Group Learning (Lectures)

Monday 10:00am - 11:50am

Group Learning (Workshops)

Thursday 9:00am – 11:50am

Independent Learning

Outside of class time

We have NO control over timing or location of classes (sorry)

We have both online and face to face group learning activities

You are expected to attend ALL lectures and tutorials

Group Learning (Lectures)

- Commence in week 1.
- Used to introduce you to the key *knowledge* in this course.
- Will include active learning strategies to encourage engagement between staff and students

Group Learning (Workshop)

- Commence in week 1.
- Provide an opportunity to (i) apply the *knowledge* gained in lectures and independent learning, and (ii) develop the *skills* necessary to successfully complete course assessments.
- Active participation is expected.
- Presentation assessment will occur in this slot in week 10-12

Independent Learning

- Weekly activities to (i) develop your lifelong learning skills and (ii) support your preparation for the course assessments

Course Timetable

DENT1020 Dental Science I Timetable - 2026

Week	Week commencing (Monday)	Module	Lecture	Workshop	
			Monday 10-12 noon Auditorium Room 4401 Level 4	Thursday 9-12 noon Auditorium Room 4401 Level 4	
1	23-Feb	1	Introduction to course; Terminology; Introduction to anatomy, skull and paraoral tissues	No workshop	
2	2-Mar	1 & 2	Cells and cellular processes	Small group activity: terminology and cellular biology	
3	9-Mar	2 & 2	Embryology of head, face and oral cavity	Library Workshop (Lars Eriksson)	
4	16-Mar	2	Embryology of head, face and oral cavity	3D visualisation workshops Embryology	
5	23-Mar	2	Tooth development, eruption and shedding	Age estimation workshops using OPGs	
6	30-Mar	1 & 2	Structure and function of bone (MP)	Dental histology workshop 1 (Bone)	Written commentary due
	6-Apr		Mid-semester break	Mid-semester break	
7	13-Apr	3	Soft tissue structures of the orofacial region	Dental histology workshop 2 (Soft Tissue)	Saturday 18 April Mid Sem requested
8	20-Apr	3	Epithelium , connective tissue and lymph (MP)	Dental histology workshop 3/ Infographic group work	
9	27-Apr	4 & 5	Enamel, Dentine-pulp complex and periodontium (MP)	Dental histology workshop 4/ infographic group work	
10	4-May	4 & 5	Public Holiday	Introductory Physics and Chemistry (Submit infographics in class)	Infographic submission due
11	11-May	All	Week 11: Health Infographic presentations (Panel: SS/SH)	Week 11: Health Infographic presentations (Panel: SS/JZ)	Presentations
12	18-May	4	Saliva composition and salivary glands	Development of oral microflora (JS)	
13	25-May	All	End of semester information session, review and mock exam	Q&A drop in session for exam review	
	SWOTVAC/ Exam block				

Assessment summary

Category	Assessment task	Weight	Due date
Examination	In-Semester Exam	30%	In-semester Saturday
	 Identity Verified		28/03/2026 - 2/05/2026
	 In-person		
Paper/ Report/ Annotation, Essay/ Critique	Written Commentary	10%	13/04/2026 5:00 pm
	 Online		
Creative Production/ Exhibition, Performance, Presentation, Poster	Health Infographic Presentation (Group)	20%	7/05/2026 - 14/05/2026
	 Identity Verified		Specific presentation rosters will be provided by Week 6 on Blackboard. Infographics should be submitted one week before the presentation date allocated to the group.
	 Team or group-based		
	 In-person		
Examination	End of Semester Exam	40%	End of Semester Exam Period
		MUST PASS	6/06/2026 - 20/06/2026
	 Hurdle		
	 Identity Verified		
	 In-person		

Check ECP and blackboard folders under Assessment.

Marking rubrics and exemplars will be available for Assessment 2 and 3 soon.

Accessing Learning Management System

<https://elearning.uq.edu.au/student-guides-ultra/getting-started-learnuq-ultra-student>

<https://echo360.net.au/media/314da199-9dfb-4134-95f7-8a8d6b5d8e28/public>

Access your Learn.UQ courses

You can access your current courses and any completed within the last year.

1. Go to [Learn.UQ](#).
2. Log in with your student ID and password.

Group Feedback (as a whole cohort)

- Mid-semester exam
- Infographic presentation assessment task
- End-of semester exam paper (after release of course marks)

Individual Feedback

- Mid-semester exam paper (instructions to request paper viewing TBA)
- End-of semester exam paper (instructions to request paper viewing TBA)
- Written Commentary assessment task

Fehrenbach MJ, Popowics T. Illustrated Dental Embryology, Histology, and Anatomy. 5th edition. Missouri, USA: Elsevier Saunders, 2020.

Fehrenbach MJ. Illustrated Dental Embryology, Histology, and Anatomy (Student Workbook), 5th edition. Missouri, USA: Elsevier Saunders, 2020.

Brand RW, Isselhard DE, Satin E. Anatomy of Orofacial Structures: A Comprehensive Approach. Enhanced 8th edition. St Louis, Missouri: Elsevier Health Sciences, 2019.

Lee G. Microbiology and Infection Control for Health Professionals, 6th edition. Melbourne, Victoria: Pearson Australia.

Nanci A. Ten Cate's Oral Histology: Development, Structure and Function. 9th edition. St Louis, Missouri: Elsevier, 2017.

You are responsible for checking the DENT1020 Blackboard site daily, for any updates and notifications relating to the course

Lecture slides will be uploaded to the Blackboard site the day before each lecture

Tutorial resources and independent learning resources will be uploaded to the Blackboard site on a regular basis – a few days before the in person session where they will be reviewed.

Blackboard announcements are the main place I will announce any important information to students

If it's on Blackboard, it's examinable!

URL: <http://learn.uq.edu.au>



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Questions?





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Thank you

Dr Sowmya Shetty

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